

Christophe de Dinechin

Software as a Passion

LANGUAGES | SYSTEMS | GAMES | PHYSICS | ART | TOOLS

◈ ABOUT

Pushing the boundaries by writing highly innovative and differentiating software to make as much of a difference as possible.

French software developer — games, languages, operating systems

Amateur physics researcher

Tinkerer in strange forms of art

◎ KEY SKILLS

- ✓ Languages & Compilers
- ✓ Systems & Virtualization
- ✓ Games & 3D
- ✓ Physics
- ✓ Art & Culture
- ✓ Tools & Apps

▤ TECHNOLOGIES

C++ · C · RPL · Lisp · XL · Tao3D · ELFE · Linux · macOS · Windows · SPICE · Emacs · OpenShift · Make

i INFO

★ Tinkerer in strange forms of art

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⌘ github.com/c3d

◉ c3d.github.io

Scan the constellation
https://c3d.github.io/cv/

LANGUAGES & COMPILERS

HP aC++ Compiler & C++ ABI — 1998-2000

- Worked on C++ templates and exception handling
- Initiated / contributed to the common C++ ABI for Itanium

DB48X - Modern RPL Calculator — 2022-Present

- Rebuild and improve the HP48-family user experience
- Rich RPL data types and built-in functions with Lisp-level programming power

SYSTEMS & VIRTUALIZATION

DxO ONE Connected Camera — 2015-2016

- Dual-core ARM + ISP real-time system
- Improving software quality on a highly capable connected camera

HPVM - Enterprise Virtualization — 2000-2010

- Started the HP Integrity Virtual Machines project in late 2000
- Early bring-up with Todd Kjos and Jonathan Ross (4-way HP-UX guests)

GAMES & 3D

Alpha Waves / Continuum — 1989-1990

- Started in 1988, published by Infogrames in 1990
- First 3D platform game; true 6-axis 3D with physics-like interaction

Firefly

- Real-time ray-marching game experiment with distance-field physics
- Collisions computed by the same GLSL distance field as rendering

PHYSICS

A Theory of Incomplete Measurements

- A Theory of Incomplete Measurements — unifying GR and quantum mechanics
- New relativity principle: do physics with any instrument

Réunifions la physique

- Gentle French introduction to incomplete-measurement physics
- Companion to A Theory of Incomplete Measurements

ART & CULTURE

3DIVART

- Adapt ray-tracing / ray-marching shaders for auto-stereoscopic screens via Tao3D
- Synthetic 3D art you can see without glasses on suitable displays

Informagie

- Started as a student around the same time as Alpha Waves
- Finished after 32 years and published

TOOLS & APPS

CapaVoit

- First Mac application, built at Axlog Ingénierie in Paris
- MacDraw-like editor with a built-in numerical solver

HP DE200C MythTV

- HP DE200C digital entertainment appliance (Linux PC in a CD-player shell)
- Converted with MythTV, DVD drive, more RAM, and a TV tuner